Development of a low-cost portable scale integrated into a smartphone case based on image processing force estimations

Mohammad Zeshaan Shahid [1], Dalia Osman [1], and Yohan Noh [2]

[1]Affiliation not available
[2]Brunel University London

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Abstract

This paper presents the development and implementation of a low-cost, portable scale utilizing a smartphone and image processing technology. The main objective of this research is to explore the potential of using a compliant mechanism, markers, and image processing techniques to accurately determine the weight of an object. We designed a 3D printed platform fitted with a webcam to capture images of white markers on a black surface. The Image Processing Toolbox on MATLAB was used to detect two parameters, the number of white pixels and the centroid positions of the white pixels with respect to the centre of the image in each marker. Calibration is carried out to convert these two parameters into weight, hence accurately estimating the weight of an object placed on the platform.

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Development of a low-cost portable scale integrated into a smartphone case based on image processing force estimations.

Mohammad Zeshaan Shahid[1], Dalia Osman[1*] and Yohan Noh[1*]

[1] *Dept. of Mechanical and Aerospace Engineering, Brunel University London, United Kingdom*

* *IEEE Member*

Corresponding author: Yohan Noh (e-mail: Yohan.noh@brunel.ac.uk).

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Abstract—This paper presents the development and implementation of a low-cost, portable scale utilizing a smartphone and image processing technology. The main objective of this research is to explore the potential of using a compliant mechanism, markers, and image processing techniques to accurately determine the weight of an object. We designed a 3D printed platform fitted with a webcam to capture images of white markers on a black surface. The Image Processing Toolbox on MATLAB was used to detect two parameters, the number of white pixels and the centroid positions of the white pixels with respect to the centre of the image in each marker. Calibration is carried out to convert these two parameters into weight, hence accurately estimating the weight of an object placed on the platform.

Index Terms—Image-processing, weight estimation, smartphone-based scale, 3D-printed platform

I. INTRODUCTION

Weight measurement of objects is integral to various applications, ranging from domestic to industrial uses. Traditional weighing scales, although useful, can be bulky, expensive, and require regular maintenance and calibration. They are relatively big, so they cannot always be carried with users, for example in a bag or pocket, when required for rough measurements of objects in a range of daily situations. Some examples of use include measurement of the weight of everyday objects, such as food items and parcels, or in an airport check-in area, for ensuring weight of personal bags and luggage are not exceeding the weight limit. Another use is for outdoor activities such as school nature field trips, where they may weigh plants or some small specimens as part of an investigation, and a smaller portable weighing scale may become handy.

Most commercialised scales are fabricated using four compliant mechanisms with strain gauges attached onto them. The strain gauges can measure their deformations when subjected to external loads, and the measured deformation can be used to estimate the external loads [1]. On the contrary to the conventional approach, this paper discusses the development of a low-cost, portable scale utilising an image processing technology.

Currently, image processing techniques have been implemented in developing scales, tactile and force/torque sensors, to detect an object's texture, weight (force/torque) estimation, contact area, as well as for tumour localization. [2-5].

The goal of this project is to devise a portable image-based weight estimation system integrated into a smartphone (mobile phone) case consisting of its body and a foldable mechanism. The concept idea of the scale system is as follows (Fig. 1a): The smartphone case is comprised of its body, four foldable markers, and four foldable legs which are compliant structures (Fig. 1b). Ordinarily, the scale is used as a smartphone case with the four legs folded (Fig. 1c). When being used as a scale, the four legs and the four markers are unfolded (Fig. 1d), and the smartphone is placed under the case body. Recent smartphones have a high-resolution camera with the fastest FPS (frames per second) and a high-performance CPU and GPU, making

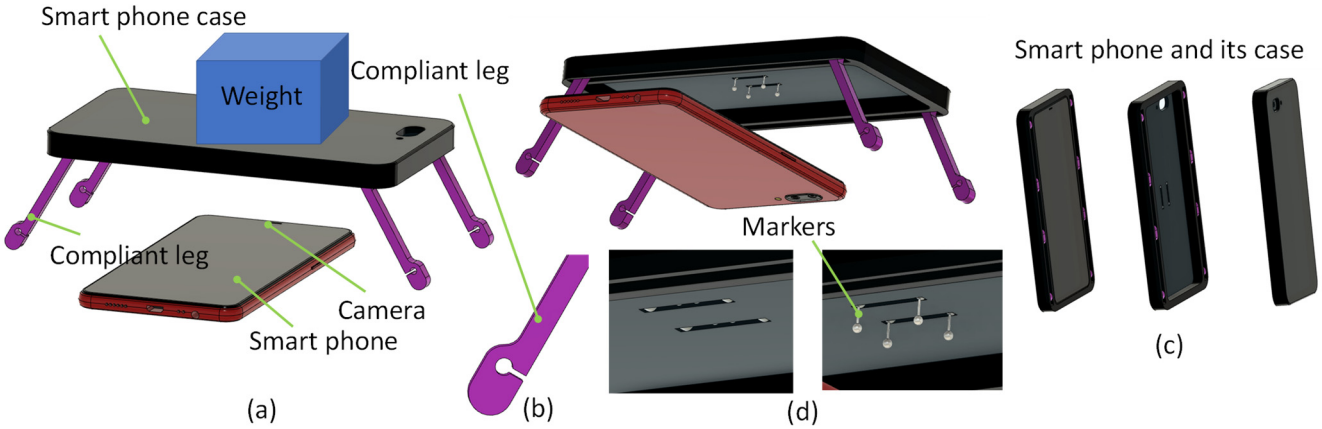

Figure 1: Conceptual idea of a portable scale with a smartphone

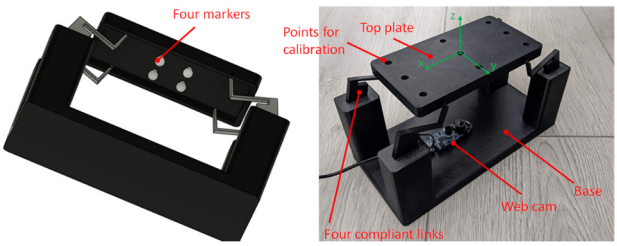

Figure 2: Overview of the design of a portable scale with a compliant mechanism, markers, a webcam.

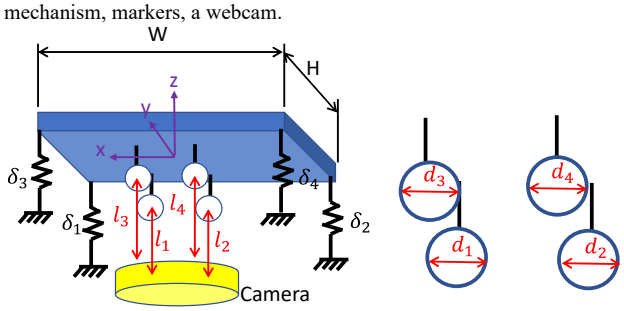

Figure 3: Overview of the sensing principle of a portable scale with a compliant mechanism and markers.

(a) (b) (c)

Figure 4: Converted binary image from grayscale image.
(a) Two parameters: markers' sizes (d_i), and measured distances (x_i, y_i)
(b) Original converted grey scale image from RGB image
(c) Binary image converted from grey scale image

image capture and processing in real time feasible at a high sampling rate. Besides, people always carry a smart phone around with them, meaning they can use the weight measuring tool anywhere and at any time.

In this paper, a conceptual scale design and its prototype will be described, and the weight measurement sensing principle, calibration, and the proposed scale's performance will be presented in more detail, although the smartphone case does not include the foldable mechanism in this prototype, and an actual smart phone was not used.

II. SYSTEM DEVELOPMENT AND IMAGE ANALYSIS

This section provides an in-depth view of the materials, methods, and image processing techniques used to create and operate the portable smartphone-based scale. It also elaborates on the design and production of the 3D-printed platform, the integration of the webcam PCB, image capture and processing, and the calibration procedure using multiple linear regression.

A. 3D-Printed Compliant Mechanism

A 3D-printed platform fabricated from ABS (acrylonitrile butadiene styrene) filament consists of four white markers, and four deformable compliant links, which were designed based on four cantilever beams as shown in Fig. 2. Under the assumption that a smart phone camera is used, a webcam is fixed on the base plate. Wherever external loads are applied on the top plate, the deformations of the four cantilever's beams can be measured. Not only the gravitational force but also two moment values such as M_x, and M_y can be estimated as shown in Eqs. (1)-(4). For this reason, four deformable compliant links designed based on four cantilever beams are attached as seen in Fig. 2.

$$f_i = k\delta_i \tag{1}$$

$$\delta_i = \frac{f_i L^3}{3EI},\ f_i = \frac{3EI}{L^3}\delta_i,\ \text{and}\ k = \frac{3EI}{L^3} \tag{2}$$

where L: Cantilever beam's length, E: Young modulus of material, I: Area moment of inertia, and k: spring coefficient

$$\delta_i(d_i, x_i, y_i)\ ,\ \text{where}\ \ i = 1, 2, 3, \text{and } 4 \tag{3}$$

$$\begin{bmatrix} F_z \\ M_x \\ M_y \end{bmatrix} = \begin{bmatrix} k & k & k & k \\ k\frac{H}{2} & k\frac{H}{2} & -k\frac{H}{2} & -k\frac{H}{2} \\ k\frac{W}{2} & -k\frac{W}{2} & k\frac{W}{2} & -k\frac{W}{2} \end{bmatrix} \begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \\ \delta_4 \end{bmatrix} \tag{4}$$

When an external load is exerted onto the top plate, the links are deformed. At this point, the markers' sizes (d_i) as they appear on the image, and their measured distances (x_i, y_i) from the centre point vary, and these values can be measured from the camera's image using image processing technology as seen in Figs.3 and 4. From these values, the four deformations of the four cantilever beams can be indirectly measured, in order to get an estimation of the applied external load.

B. Vision system and image processing

We integrated a commercially available webcam, the Logitech C270, into the base of the platform, with its case removed. The camera lens was positioned above the centre of the platform and aligned with

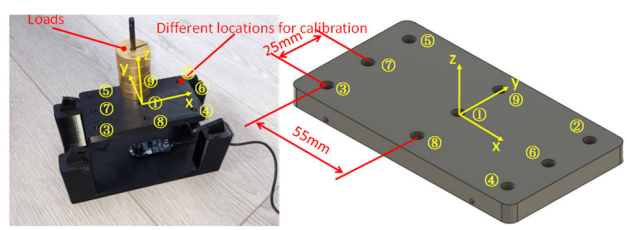

Figure 5: Overview of a sensor calibration device and its procedure.

Table I: Experimental Configuration of the Applied Loads for Calibration

Positions on calibration base (see Fig. 5)	Force/ Moment components	Force and Moment ranges
1	F_z	-4.9N to 0N
2	F_z, M_x, M_y	-4.90N to 0N, -12.26Ncm to 0Ncm, 0Ncm to 26.97Ncm
3	F_z, M_x, M_y	-4.90N to 0N, 0Ncm to 12.26Ncm -26.97Ncm to 0Ncm
4	F_z, M_x, M_y	-4.90N to 0N, 0Ncm to 12.26Ncm 0Ncm to 26.97Ncm
5	F_z, M_x, M_y	-4.90N to 0N, -12.26Ncm to 0Ncm -26.97Ncm to 0Ncm

the four white markers on the black surface (Figs. 2, 3 and 4). A data acquisition application created by MATLAB was used to initialize the webcam and create a user-friendly graphical interface for real-time image capture and processing. When a weighted object is placed on the top plate platform, the application captures images of the white markers on the black surface. A Gaussian filter to reduce noise is applied, and the images are converted from RGB images to grayscale [6]. Then, these grayscale images are converted into binary images by setting a certain threshold value. The binary images are divided into four equal sections (regions of interest (ROI)), each corresponding to one of the markers. The number of white pixels (d_i) in each section is counted, and the centroid positions (x_i, y_i) of the white pixels in each section with respect to the centre of the image are calculated using the MATLAB function 'regionprops.' Eventually, these values are used as input parameters to estimate the external force/torque components.

III. SENSOR CALIBRATION

A. Calibration Procedure Using Multiple Linear Regression

To convert a set of the counted number of white pixels (d_i) and the centroid positions from the processed images into physical quantities, which is force and two moment values, a calibration process is required. A series of loads (each 100g, 0.98N) were placed at different positions on the top plate ①-⑤, to apply force/torque components, and the corresponding marker size and distance data were collected from the images, as seen in Fig. 5. Mounting a series of loads at different locations enabled different conditions of force and moments on the top plate as shown in Table 1. This calibration process was repeated three times. *Multiple linear Regression* (*MLR*) was implemented to calculate a calibration matrix (C) using the known force/torque values and the mean values of the observed marker size (d_i) and distance data (x_i, y_i) obtained [7-8]. The calculated calibration matrix enabled the prediction of force/torque values based on its effect on the marker sizes and distances as shown in Eqs. (5)-(7).

$$[F] = [C] \times [X] \tag{5}$$

where C: Stiffness Coefficient Matrix; X: Pixel Input Parameters Coefficient Matrix; F: Force and Moment Components

$$\begin{bmatrix} F_z \\ M_x \\ M_y \end{bmatrix} = \begin{bmatrix} C_{1,1} & C_{1,2} & \cdots & C_{1,11} & C_{1,12} \\ C_{2,1} & C_{2,2} & \cdots & C_{2,11} & C_{3,12} \\ C_{3,1} & C_{3,2} & \cdots & C_{3,11} & C_{3,12} \end{bmatrix} \times \begin{bmatrix} P_1\ Count \\ \vdots \\ P_4\ Count \\ X_1\ Distance \\ Y_1\ Distance \\ \vdots \\ X_4\ Distance \\ Y_4\ Distance \end{bmatrix} \tag{6}$$

$$C = \begin{bmatrix} 0.967 & -4.989 & -8.805 & 5.938 & 357.216 & -6.199 & 196.438 & -583.162 & 179.471 & -211.860 & 437.636 & -16.777 \\ 0.018 & -0.129 & 0.191 & -0.093 & -5.519 & -9.868 & -0.999 & -3.020 & 2.784 & 2.154 & 10.953 & 5.718 \\ 0.277 & 0.001 & -0.311 & 0.010 & 4.356 & 1.110 & 1.546 & -10.231 & -5.456 & 7.279 & -1.956 & 5.554 \end{bmatrix} \times 10^{-3} \tag{7}$$

B. Verification of Calibration Matrix

To verify the calibration matrix, a set of loads (each 100g, 0.98N) are placed at five different positions ①, and ⑥-⑨. Mounting a series of loads at different locations, again, can apply different conditions of force and moment on the top plate as shown in Table 2.

Table II: Experimental Configuration of the Applied Loads for verification

Positions on calibration base (see Fig. 5)	Force/ Moment components	Force and Moment ranges
1	F_z	-4.9N to 0N
6	F_z, M_y	-4.90N to 0N, 0Ncm to 26.97Ncm
7	F_z, M_y	-4.90N to 0N, -26.97Ncm to 0Ncm
8	F_z, M_x	-4.90N to 0N, 0Ncm to 12.26Ncm
9	F_z, M_x	-4.90N to 0N, -12.26Ncm to 0Ncm

At this point, the calibration matrix is multiplied with the observed marker size (d_i) and distance data (x_i, y_i) from the captured images, to estimate force/torque values. The performance of the portable scale was examined by comparing the estimated force (F_z) and moment (M_x, M_y) values calculated using the calibration matrix, with the actual load that was applied, detailed in Fig. 6 and Table 2.

IV. DISCUSSION

There are some limitations associated with the proposed system. The system shows significant accuracy errors as shown in Table 3. Although the system provides reasonably accurate weight measurements, it may not be as precise as some high-end weighing scales. This is mainly caused by inaccurate force/torque ground truth values during the calibration process. As additional loads are applied on the top platform, the four legs are further deformed. When a series of loads are applied at the centre on the top platform, the ground truth force is the same as gravity force, F_z. However, when the loads are mounted onto the top plate at locations other than the centre, due to tilting, it cannot be guaranteed that the ground truth force's direction is in the gravity direction, meaning tangential components of F_x and F_y should be considered to calculate a calibration matrix. The omission of the tangential force components inevitably resulted in the significant error values. These errors could affect crosstalk, although in this paper, the performance properties of the proposed scale, such as crosstalk, nonlinearity, hysteresis, and repeatability were not evaluated due to these errors.

The system's performance also has a dependency on smartphone and camera quality. The Logitech C270 has an HD level camera (1280 x 720), although it was not enough to detect the variation of the number of pixels and distances with respect to the centre of the image when the external load was applied on the top plate. For this reason, the length of the cantilever beam had to be made longer in order to allow larger variations of these parameters, and flexible compliant material ABS (Acrylonitrile butadiene styrene) was used, although this approach resulted in large errors due to the large amount of tilting as shown in Fig. 6 and Table 3.

This accuracy issue on the compliant mechanism and camera could be resolved by adopting a high-quality smart phone with a high-resolution camera and fastest FPS whose resolution is greater than 12MP with more than 60 fps. This would detect the large variation of those parameters despite infinitesimally small deformations in the compliant mechanism. This in actual fact is believed to have a valuable effect in miniaturising the overall size of the system platform

for the smartphone case as shown in Fig. 1 for the following reasons: 1) the high-resolution camera facilities adopting a modified cantilever beam with shorter length, as well as smaller markers because it would detect larger variation of those parameters when load is applied; 2) any harder materials [9] whose stiffnesses is stronger than polycarbonate (PC), TPU (Thermoplastic polyurethane), and ABS (Acrylonitrile Butadiene Styrene), commonly being used for smartphone cases, could be used for the case body, as well as the legs, again because tiny deformations could be detected by the high-resolution camera.

V. CONCLUSION and FUTURE WORKS

In this paper, although the prototype was not a smart phone case with a foldable mechanism and foldable markers, the principle and weight measurement concept using a complaint mechanism, markers, a webcam, and image processing technology were verified. Although the camera and mechanism issue caused the weight measurement to have huge errors, adopting a high-resolution camera would not only improve the accuracy, but also make the overall size of a smart phone case compact and light as shown in Fig. 1. As future works, in designing a smart phone case with a foldable mechanism and foldable markers as shown in Fig. 1, optimising a complaint mechanism with a high-resolution camera and choosing a case and its complaint mechanism material will be a key challenge to improve the scale's performance properties such as accuracy, sensitivity, nonlinearity, repeatability, and hysteresis. In addition, different image processing algorithms will be devised to better enhance the scale's performance properties. Furthermore, depending on a smartphone's brand, it will have different hardware specifications, such as the camera resolution, the lens, and a different focal point. For this reason, adapting one calibration matrix to other mobile devices for measuring weight remains the biggest challenge.

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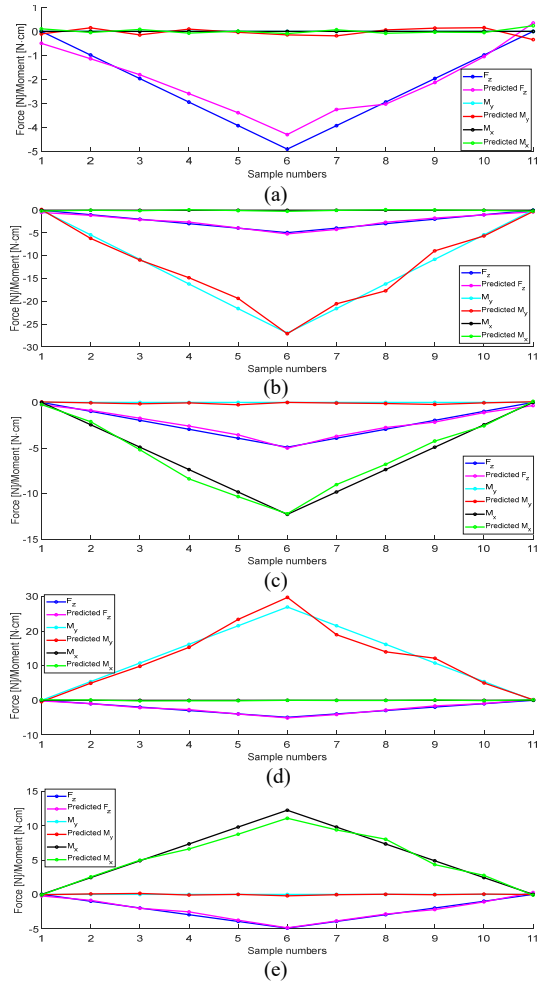

Figure 6: Comparison between ground truth values and estimated ones
(a) Comparison between real F_z and estimated F_z
(b) Comparison between real F_z, -M_y and estimated F_z, -M_y
(c) Comparison between real F_z, -M_x and estimated F_z, -M_x
(d) Comparison between real F_z, +M_y and estimated F_z, +M_y
(e) Comparison between real F_z, +M_x and estimated F_z, +M_x

Table III: Sensor Error Property

Force / Moment	Range	Maximum Error
F_z	-4.90N to 0N	18.08%
M_y	-26.97Ncm to 26.97Ncm	17.29%
M_x	-12.26Ncm to 12.26Ncm	13.83%

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